

Object-Oriented Programming in Java

Object-Oriented Programming (OOP) is a paradigm based on objects and classes. Java is one of the most widely used OOP languages, focusing on reusability, modularity, and encapsulation.

Key Points

1. Class: Blueprint of an object.
2. Object: Instance of a class.
3. Inheritance: Mechanism for reusing code.
4. Polymorphism: Ability to take many forms.

Example Code

```
class Person {
    String name;
    int age;

    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    void greet() {
        System.out.println("Hello, my name is " + name);
    }
}

public class Main {
    public static void main(String[] args) {
        Person p = new Person("Alice", 30);
        p.greet();
    }
}
```

Conclusion

Java OOP principles improve software design and reusability.